Incident report analysis

Instructions

As you continue through this course, you may use this template to record your findings after completing an activity or to take notes on what you've learned about a specific tool or concept. You can also use this chart as a way to practice applying the NIST framework to different situations you encounter.

Summary	During the day, our organization experienced a possible DDoS attack to our internal network, which compromised our network operativity for 2 hours until it was resolved. The attack was noticed after several internal employees reported that our organization’s network services weren’t responding accordingly. After starting an investigation, our security team pointed out a flood of ICMP packets as the possible main cause from stopping our services, then recognised as a DDoS attack. The team decided to respond by blocking all ICMP incoming traffic, deactivating all non critical network services and getting back working all critical network services. Further investigation indicates that the malicious actor started the ICMP flood passing by an unconfigured firewall of the network, this lead as the entry point inside the network for the attack. After that, the security team implemented the right intervention in order to harden the network and prevent the continuation of the attack.
Identify	The cybersecurity team investigated and identified an unconfigured firewall as the main entry point for the malicious actor, who started a DDoS attack by flooding the internal network with ICMP packets, in order to disable many

	network services. This is known as an ICMP flood attack.
Protect	The network security team implemented several solutions in order to address this and future similar attacks. They added a new firewall rule in order to limit the volume and rate of incoming ICMP traffic. And an IDS/IPS system in order to filter out some ICMP traffic based on suspicious characteristics.
Detect	To detect future ICMP flood attacks to the network organization, the security team implemented a Network monitoring software and SIEM tools to detect and recognize abnormal traffic patterns. Even a verification of the source IP address on the firewall has been implemented, in order to identify possible spoofed IP addresses on incoming ICMP traffic. This will help to anticipate the starting of the attack and to respond immediately.
Respond	The incident management team responded by blocking ICMP packets and stopping non-critical network services offline, and restoring critical network services. For future security events, the team will isolate affected systems to prevent further disruption to the network, and will analyze network logs to check for suspicious and abnormal activity. We also informed management roles about this attack, and reported the incident to the appropriate legal authorities where applicable.
Recover	As the investigation continues by analyzing possible changes in data inside the network, all the services will be gradually reactivated, when the network environment is classified as safe. In case of a future possible DDoS attack by ICMP flooding, the following

	sequence will be followed as recovery: firstly, external ICMP flood attacks must be blocked at the firewall. After that, all non-critical network services should be stopped to reduce internal network traffic. Next, critical network services should be restored first. Finally, once the flood of ICMP packets have timed out, all non-critical network systems and services can be brought back online.

Reflections/Notes: Some extra implementation in order to strengthen the network security should be applied in order to prevent future attacks. One priority should be better security hardening over network devices, setting them up and updating them monthly, even having an updated map of all the network architecture, in order to know any possible device that can put the organization in danger. Another priority is the implementation of network segmentation in order to isolate different network environments, without compromising the entire network.